UNIVERSIDAD DE CHILE

Andres Gonzalez <andres.gonzalez.s@ug.uchile.cl>

Request for R/SAS codes: AAF calculations for alcohol-related cancers

7 mensajes

Andres Gonzalez <andres.gonzalez.s@ug.uchile.cl> 15 de mayo de 2026 a las 14:21
Para: asherk@uvic.ca

Dear Dr. Sherk,

I hope this email finds you well.

My name is Andrés González, a PhD student in public health from the University of Chile. I am writing to kindly request access to the analytical codes referenced in the appendix of the Alcohol and Society 2016 report (https://arkiv.iogt.se/pdf/Appendix-to-Alcohol-and-Society-2016_Methods-used-for-estimating-alcohol-related-cancers-in-Sweden.pdf).

Specifically, we are highly interested in the R codes—originally written by Shield and Rehm for the GBD project and revised by your team to include prostate cancer—used to calculate the Alcohol-Attributable Fractions (AAFs) for cancers. Additionally, if possible, we would appreciate access to the SAS codes used to calculate the attributable deaths and hospitalizations based on these AAFs.

Our research team is currently conducting a microsimulation study in Chile. As part of our methodology, we need to accurately calculate Relative Risks (RRs), but we have encountered challenges and inconsistencies in RR estimates from other sources. Having the opportunity to review and adapt your code would be immensely helpful for our project.

We will, of course, fully acknowledge your valuable contribution and the original authors in our work and any resulting publications.

Thank you very much in advance for your time, help, and consideration.

Sincerely,

Adam Sherk <asherk@uvic.ca> 15 de mayo de 2026 a las 18:25
Para: Andres Gonzalez <andres.gonzalez.s@ug.uchile.cl>

Dear Andres,

Sure, I can provide this - although the project you reference in Sweden is quite old now and the RR functions have in some cases been updated by the newest WHO Global Status Report on Alcohol and Health or IHME Global Burden of Disease studies.

In any case, I attach the R code used for the project for which you include the link in your email. Yes, these are originally from Shield and Rehm.

I further attach the SAS files I wrote to conjoin the enumerated cancer counts with the alcohol-attributable fractions (AAFs).

Are you focusing in your microsim only on cancer? If yes, I would suggest using the RR functional forms that are available in the Shield et al publication of 2025 in Lancet Public Health.

Shield, K., Franklin, A., Wettlaufer, A., Sohi, I., Bhulabhai, M., Farkouh, E. K., ... & Rehm, J. (2025). National, regional, and global statistics on alcohol consumption and associated burden of disease 2000–20: a modelling study and comparative risk assessment. *The Lancet Public Health*, *10*(9), e751-e761.

Here is the link: https://www.thelancet.com/journals/lanpub/article/PIIS2468-2667(25)00174-4/fulltext

Then, if you look at the Supplementary Appendix, it will show the chosen RRs for all health conditions, including cancer.

Let me know if you need anything else,
Adam

Adam Sherk, PhD
Scientist, Canadian Institute for Substance Use Research, University of Victoria
Adjunct Associate Professor, School of Public Health and Social Policy

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From: Andres Gonzalez <andres.gonzalez.s@ug.uchile.cl>
Sent: Friday, May 15, 2026 11:21 AM
To: Adam Sherk <asherk@uvic.ca>
Subject: Request for R/SAS codes: AAF calculations for alcohol-related cancers

You don't often get email from andres.gonzalez.s@ug.uchile.cl. Learn why this is important

[El texto citado está oculto]

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6 adjuntos

chronicAAF.R
16K

chronicAnalysis.R
6K

chronicRR.R
24K

D1. Create mortality files from website input.sas
10K

D2. Create AAF mortality file from R output.sas
19K

D3. Merge mortality and AAF files, calculate and output.sas
20K

Andres Gonzalez <andres.gonzalez.s@ug.uchile.cl> 15 de mayo de 2026 a las 20:49
Para: Adam Sherk <asherk@uvic.ca>

Thank you so much, Dr. Adam,

In fact, I wrote to Dr. Shield some weeks ago asking for the code, but we did not receive a response from him. This was because in the Supplement to: Shield K, Manthey J, Rylett M, Probst C, Wettlaufer A, Parry CDH, Rehm J. National, regional, and global burdens of disease from 2000 to 2016 attributable to alcohol use: a comparative risk assessment study. Lancet Public Health 2020; 5: e51–61, they added the formulas. However, we found that some formulas and variance-covariance matrices we had trouble using (e.g., Pancreatitis, p. 50: VarCov(b1, b2) = [0.01127452 -0.00015821; -0.00015821 0.01762052]).

In the document you kindly shared, I could not find the RRs. For example, it seems these RRs vary substantially with alcohol dose rather than remaining constant.

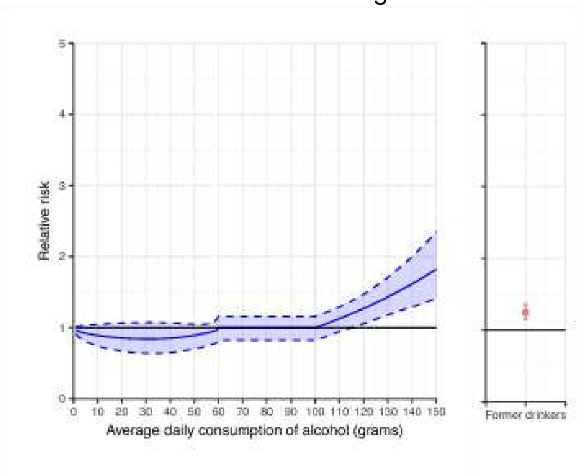

Figure S29. Relative risk for ischaemic heart disease among males 65 years of age and older

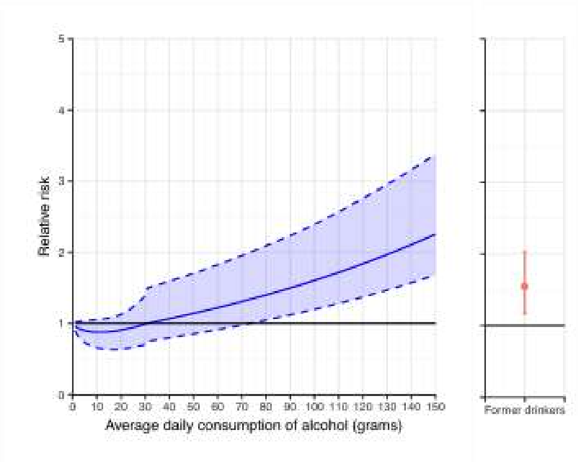

Figure S30. Relative risk for ischaemic heart disease among females 65 years of age and older

In the previous supplemental material (Shield K, Manthey J, Rylett M, Probst C, Wettlaufer A, Parry CDH, Rehm J. National, regional, and global burdens of disease from 2000 to 2016 attributable to alcohol use: a comparative risk assessment study. Lancet Public Health 2020; 5: e51–61) I was able to reconstruct the dose-specific relative risks (RRs). However, for several diseases, the reconstructed RRs appear implausible or inconsistent with expected dose-response patterns, and we have been unable to reproduce plausible results for those cases.

Ischaemic heart disease (35 to 64 years of age)	
Sex:	Male
Category:	Noncommunicable diseases / Cardiovascular diseases
ICD-10 codes:	I20–25
GHE 2015 cause category:	II.H.3
Causality:	Mukamal & Rimm, 2001; Collins *et al.*, 2009; Roerecke & Rehm, 2014 [45-47]

	Current drinkers	**Former drinkers**

Relative risk function or estimate	If(x < 60) $RR_{CD} = \exp(\beta_1 \cdot (\beta_2 \cdot \sqrt{y_1} + \beta_3 \cdot y_1^3))$
	If(60 ≤ x < 100) $RR_{CD} = 0.04571551 + \exp(\beta_1 \cdot (\beta_2 \cdot \sqrt{y_2} + \beta_3 \cdot y_2^3))$
	If(100 ≤ x) $RR_{CD} = \exp(\beta_4 \cdot (x - 100)) - 1 + 0.04571551 + \exp(\beta_1 \cdot (\beta_2 \cdot \sqrt{y_1} + \beta_3 \cdot y_1^3))$
	$where\ y_1 = \frac{x + 0.0099999997764826}{100}$
	$y_2 = \frac{60 + 0.0099999997764826}{100}$
	$\beta_1 = 0.757104$
	$\beta_2 = -0.4870068$
	$\beta_3 = 1.550984$
	$\beta_4 = 0.012$
		$RR_{FD} = 1.25$

Variance	Variance co-variance matrix (β_1, β_2, β_3, β_4): $\begin{array}{cccc} 0.107981 & 0 & 0 & 0 \\ 0 & 0.298177 & 0 & 0 \\ 0 & 0 & 0.2822218^2 & 0 \\ 0 & 0 & 0 & 0.0025^2 \end{array}$	Variance($\ln(RR_{FD})$) = 0.0427864176971434^2
Comments	The relative risk is 1 for all people, who consume on average less then 60g of pure alcohol per day, but have at least one heavy drinking occasion per month [45,48]	
Source	Rehm et al., 2016 [49] based on Roerecke & Rehm, 2012 [50]	Roerecke & Rehm, 2011 [51]

x = grams of alcohol consumed per day among current drinkers

If possible, and if it would not be too much trouble, could you or another co-author point us to the code or any additional notes that clarify how those RRs were derived? We would be very grateful. This work is particularly important for us because Chile is among the Latin American countries with the highest alcohol consumption and is a significant wine producer, so accurate local estimates are urgently needed for public health planning.

Thank you for your guidance, and I hope you have a great weekend.

All the best,
[El texto citado está oculto]

Adam Sherk <asherk@uvic.ca> 18 de mayo de 2026 a las 19:29
Para: Andres Gonzalez <andres.gonzalez.s@ug.uchile.cl>

Dear Andres,

Accept my apologies for the incorrect link I sent below. Well, not incorrect but I did not realize that the RR functions themselves did not appear in the LPH 2025 supplement that I linked below.

I am not a co-author of that article (just so you know). But, I know this work fairly well.

In December 2025, myself and another colleague received R code from Dr. Kevin Shield. The R code represents the functions used in the 2025 Lancet Public Health article that I reference in my email below and also in the WHO 2024 Global Status Report on Alcohol and Health and Treatment of Substance Use Disorders (WHO 2024 GSRAHTSUD). Long acronym!

I'm hoping this will help you find every single RR function (for current drinkers) and RR point estimate (for former drinkers) that you need for your study.

To your point about the 2020 LPHL Supplement. There are sometimes small mistakes in these supplements (e.g. copy and paste errors), since there are lot of formulas. So it's not surprising you found a few discrepancies (e.g. pancreatitis like you mention).

All the best and let me know if you need anything else,
Adam

From: Andres Gonzalez <andres.gonzalez.s@ug.uchile.cl>
Sent: Friday, May 15, 2026 5:49 PM
To: Adam Sherk <asherk@uvic.ca>
Subject: Re: Request for R/SAS codes: AAF calculations for alcohol-related cancers

[El texto citado está oculto]

Re_ R code from WHO 2024 GSRAHTSUD.zip
11K

Andres Gonzalez <andres.gonzalez.s@ug.uchile.cl> 19 de mayo de 2026 a las 11:57
Para: jose.ruiztaglem@gmail.com

Hoal José,

Voy a probar usando estos códigos que me envió Adam
[El texto citado está oculto]

Re_ R code from WHO 2024 GSRAHTSUD.zip
11K

Andres Gonzalez <andres.gonzalez.s@ug.uchile.cl> 19 de mayo de 2026 a las 11:58
Para: jose.ruiztaglem@gmail.com

FYI

---------- Forwarded message ---------
De: **Adam Sherk** <asherk@uvic.ca>
[El texto citado está oculto]
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20K

Andres Gonzalez <andres.gonzalez.s@ug.uchile.cl> 19 de mayo de 2026 a las 11:59
Para: Adam Sherk <asherk@uvic.ca>

Dear Adam,

Thank you for sending the R code and for your explanation regarding the RR functions. I appreciate your insights and the clarification about the 2020 LPHL Supplement. I'll definitely use the code you provided to enhance our approach.

All the best,
Andres
[El texto citado está oculto]